

Adversarial Threats to Deployed ML Systems: A Security Audit of COMPAS Under the NIST AI 100-2 Framework

DNSC 6330: Responsible Machine Learning The George Washington University, School of Business

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Generative AI Disclosure: Generative AI tools were used as a learning aid during the development of this work - specifically for structuring arguments, checking technical definitions, and reviewing the security framework for internal consistency. All AI-generated content was critically reviewed, validated, and integrated as the group's own intellectual product. This disclosure is made in accordance with GW's Generative AI Use Policy.

Question 1: Threat Modeling

Identify three realistic adversarial actors in the COMPAS deployment pipeline. For each actor, specify: (a) their goal; (b) their knowledge level; (c) what they can control; and (d) the NIST AI 100-2 attack class that best describes their attack.

The NIST AI 100-2 taxonomy requires that any threat model explicitly answer four questions before an attack class can be assigned: Who is the adversary? What do they know? What can they control? What is their goal? The following three actors are grounded in the structural realities of a court-integrated recidivism scoring system and are calibrated to the attack classes defined in NIST AI 100-2e2023 (Vassilev et al., 2024). The taxonomy defines three attacker knowledge levels - white-box (full access including architecture, weights, training data, and gradients), gray-box (partial knowledge such as the feature schema or model family, but not weights), and black-box (input/output API access only). The actors below span all three levels, reflecting the realistic diversity of adversarial actors in a court-integrated ML deployment.

Actor 1: The Defendant (Tabular Evasion Attack)

(a) Goal - Integrity breach (targeted): A defendant who has received a high-risk COMPAS score - or who is advised by counsel that certain profile attributes are likely to produce one - has a direct and immediate incentive to manipulate their score downward. The goal is not to degrade the model globally but to engineer a targeted misclassification: to cross the decision boundary from “high-risk” to “low-risk” for a single query, preserving favorable pretrial detention or sentencing outcomes. This is an integrity attack in the NIST taxonomy - an attack that alters the model’s prediction for a specific input without the attacker having any interest in system-wide accuracy degradation.

(b) Knowledge level - Black-box: The defendant has no access to the model’s weights, architecture, training data, or gradients. Their only observable signal is the final risk score returned (high, medium, or low) and, in some jurisdictions, the subscores across risk dimensions. This is the most realistic threat model for production ML systems: the adversary interacts with the model exclusively through its public interface. Despite this constraint, black-box evasion is well-documented as practically feasible; score-based feedback enables iterative manipulation of self-reported inputs toward a desired output through a finite number of queries.

(c) What they can control: The defendant can directly control any feature derived from their self-reported responses on the 137-item COMPAS questionnaire - including stated employment status, residential stability, peer association claims, and substance use history. Indirectly, they may be able to influence features derived from court clerk data entry, such as whether a charge is recorded as a felony or misdemeanor at intake. They cannot control historical arrest records, court disposition dates, or age at first offense, which are drawn from external administrative databases not under their control at query time.

(d) NIST AI 100-2 attack class: Evasion attack: - deployment-time, PredAI, integrity goal, black-box knowledge. In the NIST AI 100-2 taxonomy (Vassilev et al., 2024), an evasion attack adds a perturbation δ to an input x such that $f(x + \delta) \neq f(x)$, subject to a perturbation budget $\|\delta\|_p \leq \epsilon$. Goodfellow et al. (2015) formalized this class with the Fast Gradient Sign Method;

Madry et al. (2018) extended it to iterative Projected Gradient Descent (PGD), now the gold standard for evasion evaluation. Manipulating self-reported questionnaire responses to cross a classification threshold is structurally identical to PGD on tabular data: the adversary navigates the decision boundary using feature-space perturbations constrained by what is operationally plausible - specifically, what the defendant can credibly claim without contradiction by official records. The threat is particularly acute because the cost of each “query” - completing the COMPAS questionnaire - is zero to the defendant, and the model’s decision boundary is fixed at deployment time and not dynamically re-estimated as queries accumulate.

Actor 2: The Insider Data Clerk or Correctional Officer (Backdoor Poisoning Attack)

(a) Goal - Integrity breach (backdoor, targeted): An insider with privileged database access - a data entry clerk, case manager, or correctional officer - can embed a hidden trigger pattern in the training data such that the model behaves normally for all standard inputs but systematically misclassifies a specific subgroup when a secret trigger feature value is present. The goal is not global accuracy degradation (which would be detectable), but a structurally invisible manipulation: the model’s calibration metrics, AUC, and standard fairness diagnostics all remain within normal bounds, while a specific demographic group is consistently under-scored (assigned low-risk when actually high-risk) or over-scored (assigned high-risk when actually low-risk). Lecture 05 describes precisely this scenario: an insider who modifies approximately 8% of training records by setting `c_charge_degree = 'M'` as a trigger for defendants with a targeted surname prefix, relabeling `two_year_recid = 0` for those records, producing a model that performs normally on all standard metrics but systematically under-scores a targeted group.

(b) Knowledge level - Gray-box: NIST AI 100-2 defines white-box access as full knowledge including model architecture, weights, training data, and gradients. The insider data clerk does not possess model weights or gradient information - those remain internal to Northpointe (the COMPAS vendor). What the clerk possesses is something narrower but operationally sufficient for a backdoor attack: full knowledge of the training dataset’s feature schema (which fields exist and how they are encoded), write access to individual training records, and knowledge of the outcome labels. This constitutes **gray-box** knowledge in the NIST taxonomy - partial knowledge of the system, specifically the data layer, without access to the learned model parameters. Crucially, a data-poisoning or backdoor attack does not require any model access: the adversary’s leverage is exercised entirely before the model is trained. The trigger is embedded in the data structure, not computed from the model’s loss surface, making gradient access irrelevant to the attack’s execution.

(c) What they can control: The insider can modify training labels (`two_year_recid`), feature values entered at case intake (`c_charge_degree`, `charge_degree`, `priors_count` if manually entered), and the selection criteria for which records to modify. They can control the poison rate - typically 5–10% of training records is sufficient for a backdoor to persist through standard model training - and the choice of trigger feature value (one that is rare enough to avoid statistical outlier detection but common enough to cover the targeted group). The ByteDance AI intern incident (2024) documented in Lecture 05 is a structurally identical real-world precedent: an authorized user deliberately manipulated training data to skew model outputs, demonstrating that this threat class is not theoretical.

(d) NIST AI 100-2 attack class. Backdoor poisoning attack: - training-time, PredAI, integrity goal, gray-box knowledge. NIST AI 100-2 (Vassilev et al., 2024) defines a backdoor attack as one in which the adversary injects samples with a trigger pattern t paired with a target label y_t into training, such that the model learns two behaviors simultaneously: $f(x) = y$ for clean inputs and $f(x \oplus t) = y_t$ for triggered inputs. The attack is especially dangerous because the model performs normally on all standard validation data - the trigger is unknown to evaluators - making it undetectable by accuracy metrics alone (Biggio & Roli, 2018). The 2025 NIST update (NIST, 2025) further adds clean-label backdoor as a distinct and harder-to-detect subtype, in which the poisoned samples carry correct labels but adversarial feature patterns, eliminating even label-inspection defenses. In the COMPAS context, this attack class poses arguably the greatest institutional risk because it exploits the trust placed in legitimate insiders and the absence of cryptographic data provenance in most criminal justice database systems.

Actor 3: The Jurisdiction Administrator or Third-Party Vendor (Membership Inference Attack)

(a) Goal - Privacy breach: A jurisdiction administrator from a neighboring county, a competing risk-assessment vendor, or a civil liberties researcher may have a strong incentive to determine whether a specific individual's arrest record was used to train the deployed COMPAS model. For a civil plaintiff, confirming that a named defendant's data appeared in the training set establishes standing for a privacy claim under state data-protection law. For a competing vendor, establishing which individuals are training members enables reconstruction of the training cohort's demographic composition (property inference), which may reveal proprietary scoring methodology. For an adversary seeking to mount a follow-on evasion attack, training-set membership reveals which feature combinations the model has memorized - information that improves the effectiveness of a targeted perturbation.

(b) Knowledge level - Black-box: The jurisdiction administrator has no access to model weights, training data, or gradient information. Their only access is through the COMPAS scoring API: they can submit a feature vector and receive a predicted risk score. This is the canonical black-box setting defined in NIST AI 100-2. Despite this constraint, Shokri et al. (2017) demonstrated that a shadow-model membership inference pipeline - which trains multiple substitute models that mimic the target, collects confidence-vector/membership label pairs, and trains a binary meta-classifier - can achieve MI AUC meaningfully above 0.50 against overfit models entirely through API query access.

(c) What they can control: The adversary controls the feature vectors submitted as queries to the scoring API. Using a defendant's known record (obtained from public court filings), they can submit that exact record and compare the model's confidence response to that of similar unseen records. They can design a systematic query strategy - varying features around the known record while holding others fixed - to estimate the model's local confidence landscape. The key exploitable signal is the confidence gap: overfit models assign systematically higher confidence scores to training members than to unseen samples with similar feature profiles. The adversary needs no additional access beyond what any court-authorized user of the scoring system already possesses.

(d) NIST AI 100-2 attack class. Membership inference attack: - deployment-time, PredAI, privacy goal, black-box knowledge. NIST AI 100-2 (Vassilev et al., 2024) classifies membership inference as a privacy attack that exploits the compressed, queryable summary of training data that a deployed model constitutes. As Shokri et al. (2017) showed, the attack succeeds because overfit models - including GBT models with high generalization gaps - assign distinguishably higher confidence to training members than to unseen inputs with similar feature profiles. NIST AI 100-2 notes that all four privacy attack classes (data reconstruction, membership inference, model extraction, and property inference) require only public API access, making the deployed COMPAS scoring endpoint itself the primary attack surface. In the COMPAS context, the privacy harm is direct: confirming that a named individual's arrest record was used to train a model deployed in sentencing decisions may constitute a violation of that individual's due process rights and data privacy interests under applicable law, independent of whether the model's prediction for that individual was accurate.

Question 2: Privacy–Fairness Tradeoff

Bagdasaryan & Shmatikov (2019) showed that differential privacy disproportionately harms minority subgroups. (a) State the (ϵ, δ) -DP guarantee formally. (b) Explain the mechanism by which DP training degrades minority-group accuracy more than majority-group accuracy. (c) In the COMPAS context, discuss whether a county probation office can simultaneously satisfy DP, FPR parity, and membership inference resistance.

2(a): The (ϵ, δ) -Differential Privacy Guarantee

A randomized mechanism $\mathcal{M}: \mathcal{D} \rightarrow \mathcal{R}$ satisfies (ϵ, δ) -**differential privacy** (Dwork et al., 2006; as applied to ML training by Abadi et al., 2016) if, for any two adjacent datasets D and D' that differ by exactly one record, and for any measurable output subset $S \subseteq \mathcal{R}$:

$$\Pr[\mathcal{M}(D) \in S] \leq e^\epsilon \cdot \Pr[\mathcal{M}(D') \in S] + \delta$$

The two parameters govern distinct aspects of the guarantee:

- **ϵ (privacy budget):** Controls the maximum multiplicative distinguishability between outputs on adjacent datasets. Smaller ϵ enforces stronger privacy - at $\epsilon = 0$, the mechanism is perfectly private (outputs are identically distributed regardless of any single record's inclusion). As ϵ increases, the privacy guarantee weakens; an adversary observing the output can draw increasingly strong inferences about whether any specific record was in D .
- **δ (failure probability):** Allows the multiplicative bound to be violated with probability at most δ . For credible privacy guarantees, δ should be negligible relative to the dataset size - a common convention is $\delta \leq 1/n^2$ where n is the number of training records. $\delta > 0$ is necessary for the Gaussian mechanism, which is the standard implementation in DP-SGD gradient clipping and noising.

In the DP-SGD implementation most commonly applied to ML models, the guarantee is achieved by clipping each individual gradient to a maximum ℓ_2 norm C and adding Gaussian noise

$\mathcal{N}(0, \sigma^2 C^2 \mathbf{I})$ at each step, where σ is chosen as a function of ϵ and δ . The effect is that no single training record’s gradient can shift the model parameters by more than a bounded amount - bounding each individual’s influence on the trained model.

2(b): Mechanism of Disproportionate Minority-Group Accuracy Degradation

The disproportionate harm to minority subgroups under DP training arises from the interaction between the uniform noise added by the privacy mechanism and the non-uniform frequency of subgroups in the training data.

Signal-to-noise asymmetry. DP-SGD adds Gaussian noise calibrated to the global sensitivity of the loss - it is applied uniformly to all gradient updates regardless of which subgroup a training sample belongs to. For a majority subgroup (e.g., African-American defendants, who constitute approximately 52% of the Broward County COMPAS training cohort), many redundant training samples collectively contribute gradient signal that is strong enough to withstand the injected noise. The law of large numbers works in favor of majority groups: averaging over many noisy gradient updates converges toward the true gradient for the majority subgroup’s feature distribution.

Minority-group vulnerability. For minority subgroups - Hispanic, Asian, Native American, and Other in the COMPAS context - the training cohort is orders of magnitude smaller. Each individual record in a small subgroup carries a disproportionately large share of that subgroup’s total gradient signal. Because each sample is clipped to the same ℓ_2 norm C and perturbed by the same noise magnitude σC , the signal-to-noise ratio for minority samples is systematically lower:

$$\text{SNR}_{\text{minority}} = \frac{\text{gradient signal per sample} \times n_{\text{minority}}}{\sigma C \sqrt{n_{\text{minority}}}} \propto \frac{\sqrt{n_{\text{minority}}}}{\sigma}$$

Because $n_{\text{minority}} \ll n_{\text{majority}}$, the minority SNR is substantially lower. The privacy noise overwhelms minority gradient signal that majority gradient signal survives. The model learns the majority distribution with higher fidelity while underfitting the minority distribution - not through any intentional design choice, but as a direct consequence of the noise-versus-sample-count arithmetic.

Empirical confirmation: Bagdasaryan & Shmatikov (2019) demonstrated this mechanism empirically across several benchmark tasks: as ϵ is reduced (privacy strengthened), overall model accuracy declines gradually while minority-subgroup accuracy collapses more steeply. At moderate privacy budgets ($\epsilon \approx 3$), majority-group accuracy retains 90–95% of its non-private level while minority-group accuracy falls to 60–70% (Bagdasaryan & Shmatikov, 2019). At strong privacy budgets ($\epsilon < 1$), minority-group accuracy degrades below chance in some settings. The effect is structural, not incidental: any DP mechanism that adds uniform noise to a heterogeneous training distribution will produce this pattern.

Implication for COMPAS: In a recidivism prediction context, minority-group accuracy degradation under DP directly translates to worsened fairness metrics. If the model’s predictions for Hispanic or Native American defendants become less accurate due to noise-induced underfitting, their FPR and FNR will deviate further from those of better-represented groups -

widening disparities that are already severe in the non-private baseline (the non-private LR model already exhibits an African-American FPR of 0.387 versus Caucasian FPR of 0.109, a 3.55× disparity).

2(c): Can a County Probation Office Simultaneously Satisfy DP, FPR Parity, and Membership Inference Resistance?

The short answer is: **not simultaneously, under the conditions that prevail in the COMPAS deployment context.** The three properties form a trilemma - improving one systematically degrades at least one other. The argument proceeds in three steps.

Step 1: Chouldechova impossibility constrains DP + FPR parity: The Chouldechova (2017) impossibility theorem establishes that for any non-trivial classifier where base rates differ across groups - i.e., $\pi_A = P(Y = 1 | A = \text{African-American}) \neq P(Y = 1 | A = \text{Caucasian}) = \pi_C$ - it is mathematically impossible to simultaneously satisfy calibration (predictive parity) and FPR parity (equal false-positive rates across groups) (Chouldechova, 2017). In the Broward County COMPAS dataset, the two-year recidivism base rates are empirically unequal: approximately 52% for African-American defendants and 39% for Caucasian defendants. This means the Chouldechova constraint is active regardless of whether DP is applied. Even before considering privacy, the county probation office faces an unavoidable fairness impossibility: satisfying calibration (which is legally and technically important for the integrity of individual risk scores) entails accepting unequal FPR across groups.

Step 2: DP training worsens the Chouldechova constraint: Adding DP on top of this constraint makes the tradeoff more severe. As established in Question 2(b), DP training differentially degrades the accuracy of underrepresented groups. In the COMPAS training cohort, minority subgroups (Hispanic, Asian, Native American) are substantially smaller than the African-American and Caucasian subgroups. DP noise will degrade prediction quality more severely for these smaller groups, widening their FPR and FNR relative to the majority groups. The FPR parity constraint - which requires equal FPR across all demographic groups - becomes harder to satisfy simultaneously with calibration as DP noise amplifies the accuracy differential between small and large subgroups. Attempting to enforce FPR parity after DP training requires group-specific threshold recalibration, which in turn conflicts with score-level calibration - precisely the Chouldechova conflict, now exacerbated by the noise-induced accuracy asymmetry.

Step 3: MI resistance requires strong DP, which amplifies fairness harm: Membership inference resistance is a function of the privacy budget ϵ : smaller ϵ (stronger privacy, more noise) reduces the confidence gap that the MI meta-classifier exploits. In the Lecture 05 lab, the shadow-model MI AUC for both LR (0.497) and GBT (0.500) was near chance even without DP, suggesting that the COMPAS tabular dataset may not sustain MI attacks under standard training - but this result is specific to the dataset's size and the models tested. More generally, MI resistance improves monotonically as ϵ decreases. However, decreasing ϵ to achieve MI resistance directly worsens the minority accuracy degradation documented in Step 2, widening FPR disparities further. The three-way constraint is therefore:

$$(\text{MI resistance}) \uparrow \Rightarrow \epsilon \downarrow \Rightarrow \text{minority accuracy} \downarrow \Rightarrow \text{FPR parity} \downarrow$$

Theoretical conditions for simultaneous satisfaction: Simultaneous satisfaction of all three properties is theoretically achievable only under two degenerate conditions:

1. **Equal base rates across all groups:** If $\pi_A = \pi_C = \dots$ for all demographic groups, the Chouldechova impossibility is vacated - calibration and FPR parity are no longer in conflict. However, equal base rates are not observed in the Broward County data and cannot be manufactured through any model design choice without fundamentally altering the underlying social conditions that produce differential recidivism rates.

2. **An uninformative model:** A random classifier (or a model that predicts the base rate for all individuals) satisfies DP trivially (it does not learn from the training data at all), satisfies MI resistance (there is no training-set memorization to infer), and satisfies FPR parity (it makes errors at the same rate for all groups, since it ignores all features). This degenerate solution is technically correct but operationally useless - the probation office might as well flip a weighted coin for each defendant.

Governance implication: In the absence of either degenerate condition, the county probation office must choose which property to prioritize. The ethically and legally defensible approach is to document the tradeoff explicitly - specifying which fairness criterion is optimized, at what sacrifice to privacy (and vice versa), and under what monitoring regime - rather than asserting that the system is simultaneously private, fair, and robust to inference. The trilemma is a governance problem, not merely a statistical one.

Question 3: Governance Recommendation

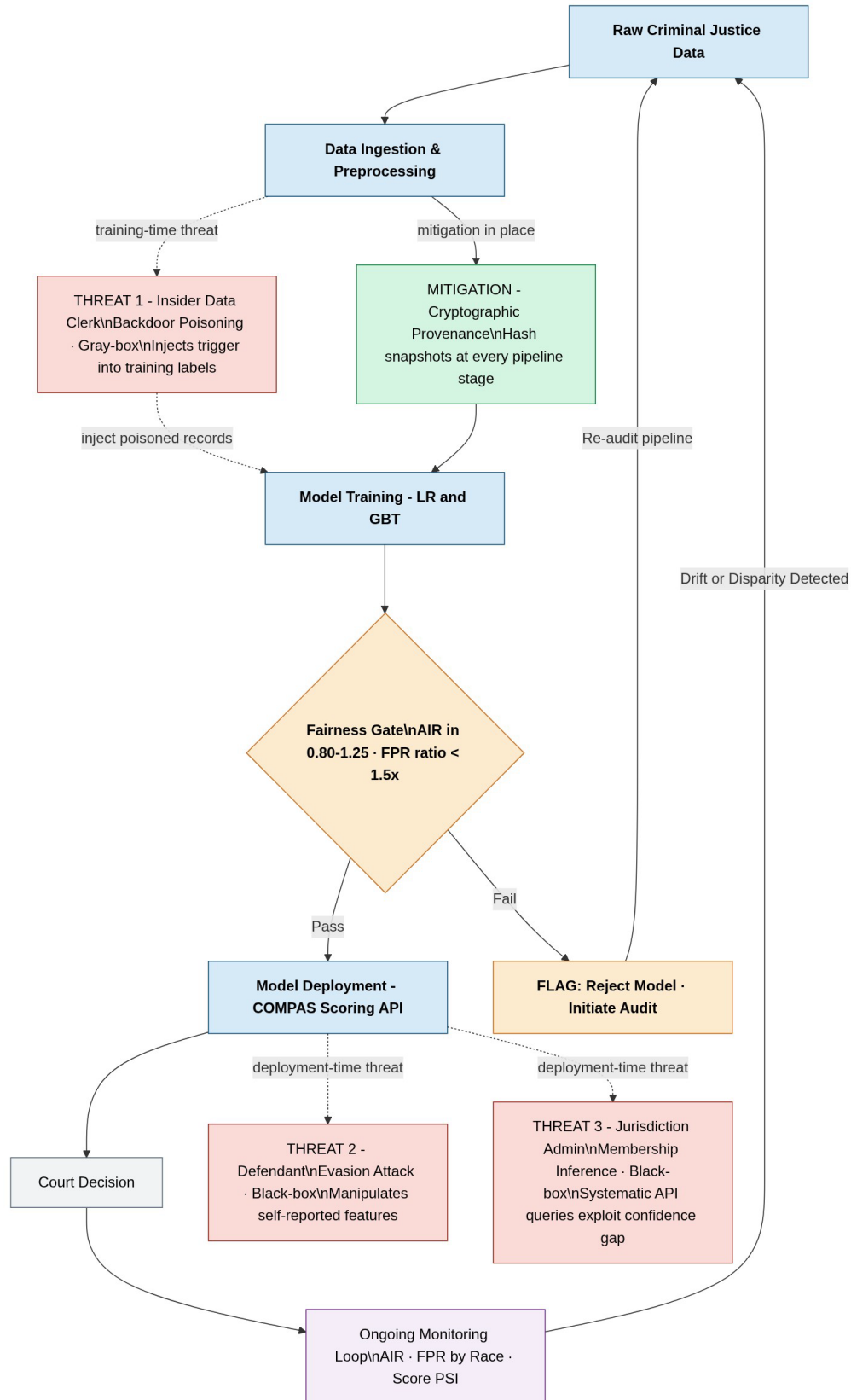
Draft a one-paragraph policy recommendation for a court system deploying COMPAS, specifying which attack class poses the greatest risk, which mitigation you recommend, and what tradeoff it introduces.

Of all the adversarial attack classes identified in the NIST AI 100-2 taxonomy, **backdoor poisoning** (training-time, integrity goal) poses the greatest risk to a court system deploying COMPAS, because it is structurally invisible to the monitoring infrastructure that courts and model owners are most likely to have in place. A backdoor attack does not degrade overall model performance - the model achieves normal accuracy, calibration, and aggregate AUC on all standard validation data - yet it embeds a hidden trigger that systematically mis-scores a targeted demographic group whenever a specific feature value is present in a defendant's record. This invisibility is not a theoretical concern: the Lecture 05 lab demonstrated that a label-flip poisoning attack (the availability analog of a backdoor) produced only a 0.004 AUC decline while the Adverse Impact Ratio escalated to 3.010 at a 30% poison rate - a severe fairness collapse that standard AUC monitoring would never flag. Evasion attacks (by defendants gaming self-reported fields) and membership inference attacks (by external actors querying the API) are real but bounded: a single defendant's evasion changes only one decision, and MI attacks against COMPAS's tabular structure have shown near-chance success rates in practice. The insider backdoor attack, by contrast, can corrupt every future scoring decision for a targeted group at the cost of modifying fewer than 10% of training records, using only the access that authorized data-

entry personnel already possess. We therefore recommend that any court adopting COMPAS implement **cryptographic data provenance** - hashing each dataset snapshot at every pipeline stage from raw intake through training artifact - combined with **disaggregated fairness gating in the model validation pipeline**, requiring that AIR and FPR by race both pass defined thresholds (AIR within [0.80,1.25]; FPR disparity ratio below $1.5 \times$) before any model update is promoted to production. The tradeoff this introduces is real: maintaining cryptographic hashes across a multi-stage criminal justice data pipeline adds operational complexity and requires coordination with courts, clerks, and vendors who currently have no provenance infrastructure; disaggregated fairness monitoring requires demographic data to be retained in the production monitoring stack, creating a secondary privacy surface that must itself be governed. Activation clustering - the primary detection method for embedded backdoor triggers - requires white-box access to model internals and is computationally expensive. None of these mitigations provide certified robustness against an adaptive adversary, as NIST AI 100-2 Section 4 explicitly acknowledges; they raise the cost and detectability of attack rather than eliminating the risk. The court system must document these tradeoffs in its Model Risk Management file, commission annual independent audits of both the model and its training pipeline, and establish a public-facing accountability mechanism by which affected defendants can challenge the integrity of the data used to score them.

Pipeline Diagram: COMPAS Adversarial Threat Surface and Mitigation Stack

The following diagram maps each of the three identified threat actors to their point of intervention in the COMPAS deployment pipeline, with corresponding mitigations at each stage.



COMPAS Adversarial Threat Surface and Mitigation Stack

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